

# Theory (Short Answers)

## 1. Mean, Median, Mode

- **Mean:** Average income of all customers.
- **Median:** Middle income value after sorting all incomes.
- **Mode:** Most frequently occurring income value.

## 2. Standard Deviation vs Variance

- **Variance:** Measures how much data is spread around the mean.
- **Standard Deviation:** Square root of variance and easier to interpret because it is in the same unit as the data.

## 3. Random Variable

A random variable is a numerical outcome of a random experiment.

**Example:** Loan Amount, Age, or Credit Score.

## 4. Conditional Probability

Conditional probability is the probability of an event occurring given that another event has already occurred.

**Example:** Probability that a customer defaults given that their credit score is low.

## 5. Bayes Theorem

Bayes Theorem is used to update probabilities using new information.

**Bank Application:** Estimating the probability of loan default based on a customer's credit score and financial history.

## 6. Empirical Probability vs Theoretical Probability

### Empirical Probability

Based on actual observations or historical data.

**Example:** 250 defaults out of 5000 loan applications.

### Theoretical Probability

Based on mathematical calculation and assumptions.

**Example:** Probability of getting Heads on a fair coin = 0.5.

## 7. Poisson Distribution

Poisson Distribution models the number of events occurring within a fixed period of time.

**Business Example:** Number of loan applications received by a bank in a day.

## **Conclusion**

These statistical concepts help banks analyze customer behavior, estimate loan default risk, make better lending decisions, and improve financial planning.